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Nickel And Its Extraction

Presented By-
1). Harpreet Khanna. (18)
2). Prince Goel (32)
3). Roopashree Panda (33)
4). Shiv Verdhan Singh (49)

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Nickel and its extraction

1. **Presented By- 1). Harpreet** Khanna. (18) 2). Prince Goel (32) 3). Roopashree Panda (33) 4). Shiv Verdhan Singh (49)
2. **Content Introduction to Nickel Extraction** of nickel Properties of Nickel Advantages Disadvantages. Classification of Nickel alloys
3. **1. INTRODUCTION TO** NICKEL • Nickel is a chemical element with symbol Ni and atomic number 28 • It is a silvery-white lustrous metal with a slight golden tinge • Nickel belongs to the transition metals and is hard and ductile • Pure native nickel is found in Earth's crust only in tiny amounts, usually in ultramafic rocks, and in the interiors of larger nickel–iron meteorites that were not exposed to oxygen when outside Earth's atmosphere
4. **1.1 OCCURRENCE • Nickel** is the earth's 22nd most abundant element and the 7th most abundant transition metal • It is a silver white crystalline metal that occurs in meteors or combined with other elements in ores • Two important groups of ores are: Laterites: Oxide or silicate ores such as garnierite, (Ni, Mg)6 Si4O10 OH)8 which are predominantly found in tropical areas such as New Caledonia, Cuba and Queensland. Sulphides:These are ores such as pentlandite, which contain about 1.5%, nickel associated with copper, cobalt and other metals. They are predominant in more temperate regions such as Canada, Russia and South Africa. Canada is the world's leading nickel producer and the Sudbury Basin of Ontario contains one of the largest nickel deposits in the world.
5. **2.EXTRACTION OF NICKEL •** The extraction of nickel and ferronickel from laterite and sulfide ores • Laterite ores occur as saprolite, smectite and limonite layers because of their different compositions and mineralogy, they require different methods of extraction ($Cu_8Ni_{16}Fe_{10}$) • Saprolite , which has a relatively low iron content, is smelted. Limonite and smectite ores, which have a high iron content, are leached and refined. ($nFe(OH)_3.16H_2O$.)
6. **FOR SULFIDES ORES [$(FeNi)_9S_8$](Pentlandite) •** Concentration, Smelting and Converting to Matte • The sulphide ores are concentrated, smelted and converted to metal-rich matte • The processing steps are as follows: (a)The valuable minerals in an ore are concentrated by froth flotation (b) This concentrate is smelted and converted into an even richer, low-iron sulphide matte
7. **PRODUCTION OF NICKEL** FROM MATTE: Nickel and cobalt are mostly recovered from low-iron matte by the following steps: (a) The matte is leached using either chlorine gas in a chloride solution, oxygen in an ammonia solution or oxygen in a solution of sulphuric acid (b) The pregnant(the solution used for heap leaching) solution is purified (c) Separate nickel and cobalt solutions are produced, usually by solvent extraction; and (d) High-purity nickel and cobalt are produced from the solutions either by electro winning or by hydrogen reduction
8. **HYDROMETALLURGICAL APPROACHES: ($Cu_8Ni_{16}Fe_{10}$)→Limonite ($FeO(OH).nH_2O$) •** Solvent Extraction – Electrowinning approach to lateritic ore beneficiation, is a hydrometallurgical method that relies on leaching, extractants, and electrowinning to produce nickel from ore Heap leaching is an industrial mining process to extract precious metals from ore via a series of chemical reactions that absorb specific minerals and then re-separates them after their division from other earth materials.

9. **PYROMETALLURGICAL APPROACHES LATERITE ($Cu_8Ni_{16}Fe_{10}$)** SAPROLITE [$nFe(OH)_3.nH_2O$] Rotary Kiln Electric Furnace method, is a pyro metallurgical approach to produce ferronickel. As a pyro metallurgical technique, this method is best suited for ores that are predominantly saprolite. • STEPS INVOLVED : Drying • Nickel laterites contain a significant amount of water, making drying an important aspect of processing. A rotary dryer is typically used to remove free moisture. Preliminary Reduction • Once ore has been dried, it is processed in a large-scale rotary kiln to remove chemically bound moisture, as well as the oxide component of the ore. This step is often referred to as pre-reduction. Reduction & Smelting • Nickel is further reduced and smelted in an electric furnace. A cooling step may be implemented after the smelting step, typically through the employment of a rotary cooler

10. **REFINING: 1.MOND PROCESS The purest** metal is obtained from nickel oxide by the Mond process , which achieves a purity of greater than 99.99%. This process has three steps: 1. Nickel oxide reacts with Syngas at 200 °C to give nickel, together with impurities including iron and cobalt. $NiO(s) + H2(g) \rightarrow Ni(s) + H2O(g)$ 2. The impure nickel reacts with carbon monoxide at 50–60 °C to form the gas nickel carbonyl, leaving the impurities as solids. $Ni(s) + 4\ CO(g) \rightarrow Ni(CO)4(g)$ 3. The mixture of nickel carbonyl and Syngas is heated to 220– 250 °C, resulting in decomposition back to nickel and carbon monoxide: $Ni(CO)4(g) \rightarrow Ni(s) + 4\ CO(g)$

11. **2.ELECTROREFINING • Nickel is** refined electrolytically from metallic nickel or nickel sulphide anodes, containing impurities iron,copper,silver,platinum,palladium etc. • The electrolyte principally contain nickel sulphate in a nearly neutral solution • Extraordinary precautions must be taken to ensure that the electrolyte at the cathode is free of copper, iron, and other impurities. • Each cathode is suspended in an individual porous walled compartment to which pure electrolyte is fed at a predetermined rate. The impure electrolyte leaving the anodes is removed from the cell and is purified by removal of iron ,copper etc. before being returned to the cathode compartments for precipitation of pure nickel on the cathode.

12. **PROPERTIES OF NICKEL PHYSICAL** PROPERTIES • Color- Silvery-White Metal • Phase- Solid • Conductivity- Fairly good conductor of heat and electricity • Ductility- It can be beaten into extremely thin sheets • Malleability- It’s capable of being shaped or bent • Luster- Exhibits a shine or glow • Hardness- Harder than iron • Ferromagnetic- Nickel is easily magnetized

13. **CHEMICAL PROPERTIES • Corrosion-** Highly resistant to rusting and corrosion • Reactivity with Oxygen- Nickel metal does not react with air under ambient conditions. • Compounds- Used for electroplating and to make nickel alloys

14. **ADVANTAGES • Nickel belongs** to transition metals. It is hard , ductile , and considered corrosion resistant because of its slow rate of oxidation at room temperature • High melting point and is magnetic at room temperature. • Low expansion

15. **DISADVANTAGES • Handling nickel** can result in symptoms of dermatitis among sensitized individuals. • High nickel concentration in air causes health issues. • High nickel concentration in soil as well as in surface water destroys plant growth and algae growth respectively

16. **CLASSIFICATIONS OF NICKEL** ALLOYS : • There are different types of nickel alloys: Nickel-copper alloys (Monels) Nickel-chromium alloys Nickel-base superalloys

17. **1.NICKEL COPPER ALLOYS** (MONELS) Ni and Cu form complete solid solution. Ni-Cu alloy contains 67%Ni and 33%Cu, called Monels. o Two types of Monels are found :- 1. MONEL METAL (METAL TREATED) : • It consists of (67% Ni + 30% Cu + small amounts of Fe, Mn, Si & C.) PROPERTIES: High strength and toughness over a range of temperature. Good weldability. Excellent corrosion resistance. Resistant to Sea Water, Chlorine

18. **APPLICATIONS: High temperature valves. Centrifugal** pump impellors & water meter parts. Steam turbine blades. 2.K-MONEL (AGE HARDENED) : It consists of 66% Ni + 29% Cu + 2.57% Al + Mn + C. PROPERTIES Wholly Non-magnetic Al increases strength & hardness above that of monel. APPLICATIONS Pump shafts, springs & valve stems, etc

19. **2.NICKEL CHROMIUM ALLOYS** Cr forms solid solution with Ni up to~30% at room temperature. High corrosion resistance is due to high Cr addition. 1. INCONEL 600: It consists of (79.5% Ni + 15.5% Cr + 8% Fe is a standard engineering alloy. PROPERTIES: High corrosion resistance at high temp. High strength and workability. Difficult to machine APPLICATIONS: Gas turbine combusters and blades. Chemical and food processing equipment. Furnace muffle & rocket skins.

20. **2. NICHROME : It** consists of 80% Ni + 20% Cr. PROPERTIES: High melting point. High electrical resistivity. Good stability from deforming. APPLICATIONS: Resistance heating coils. Rocket igniters. Jewellery casting supports. •In Ceramic manufacturing industry other Ni-Cr alloys are Inconel 601 and 625 with improved properties

21. **3.NICKEL BASE SUPERALLOYS** High temperature heat-resistance alloys, which can retain high strengths at elevated temperatures. There are three types of Ni-base superalloys:- •Ni base. •Ni- iron base. •Cobalt base. The alloys contain high Cr with Ti, Al to from precipitates and additions of Mo, Co, Nb, Zr, B, Fe. Microstructures are complex. PROPERTIES: Heat resistant and high strength at high temperature (760-980°C). Good corrosion & oxidation resistance.

22. **APPLICATIONS Aircrafts, space vehicles,** rocket engines Industrial gas turbines, high temp applications. Nuclear reactors, submarines. Steam power plants, petrochemical equipment 4.SOME OTHER NICKEL ALLOYS 1. German Silver: Consists of 60% Cu + 20% Ni + 20% Zn. PROPERTIES Hard, white & ductile. Good mechanical & corrosion resistance. APPLICATIONS Silver plated cutlery. Zippers & jewellery. Musical instruments.

23. **2. Haste alloy:- Consists** of 45% Ni + 16% Cr + 15% Mo. PROPERTIES High wear resistance. High corrosion resistance. High stress service. APPLICATIONS Bearings. Pressure vessel linings. Chemical reactor pipes.